

FRANKLIN COUNTY AP, VT, USA

WMO: 720494

Lat: 44.933N

Lon: 73.100W

Elev: 70

StdP: 100.49

Time Zone: -5.00 (NAE)

Period: 08-19

WBAN: 00152

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-22.7	-19.9	-27.5	0.3	-22.4	-24.6	0.4	-19.4	8.7	2.7	7.8	2.0	1.4	130	0.481

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.9	31.2	22.4	29.5	21.4	27.6	20.3	23.8	28.6	22.8	27.2	21.9	26.0	2.5	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.4	17.2	26.4	21.3	16.1	25.2	20.4	15.2	24.4	71.4	28.7	67.4	27.2	64.3	26.1	27.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.4	6.0	5.3	DB	-27.9	33.5	2.8	1.9	-29.9	34.8	-31.6	35.9	-33.2	37.0	-35.2	38.4	(4)
			WB	-27.8	25.3	2.8	1.4	-29.8	26.4	-31.4	27.2	-33.0	28.0	-35.0	29.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	7.9	-7.6	-5.6	-0.7	6.7	14.5	18.3	21.7	20.6	16.5	9.6	2.9	-3.4
	DBStd	11.35	7.10	6.34	6.04	5.00	4.68	3.64	3.27	2.90	4.37	4.73	5.25	6.46
	HDD10.0	2158	544	437	334	125	12	0	0	0	3	67	219	416
	HDD18.3	4107	803	670	590	351	136	43	7	11	86	273	463	674
	CDD10.0	1381	0	0	3	25	153	251	363	329	197	54	6	1
	CDD18.3	287	0	0	0	1	18	43	112	82	30	2	0	0
	CDH23.3	2222	0	0	0	15	176	344	923	541	218	5	0	0
	CDH26.7	579	0	0	0	2	44	86	289	115	43	0	0	0
Wind	WSAvg	2.0	2.2	2.2	2.4	2.6	2.3	1.9	1.6	1.6	1.7	2.0	2.3	2.0
Precipitation	PrecAvg	955	55	48	59	75	82	102	109	98	87	101	76	67
	PrecMax	1111	145	79	115	133	174	192	169	188	194	187	132	152
	PrecMin	716	24	22	18	13	31	38	50	30	44	24	25	22
	PrecStd	114	25	17	22	32	36	44	33	35	34	48	26	31
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.7	12.7	20.9	25.2	30.9	31.9	34.0	31.5	31.1	24.0	18.0	13.0
		MCWB	10.3	9.7	14.3	14.8	21.4	22.4	24.6	22.8	23.1	18.2	13.6	10.3
	2%	DB	7.5	7.7	13.1	21.6	27.5	29.1	32.0	29.9	27.6	21.3	15.3	9.8
		MCWB	5.7	4.6	8.3	12.9	17.8	20.1	23.0	21.9	21.2	16.5	10.7	7.7
	5%	DB	4.3	5.0	10.9	18.0	25.0	27.0	30.1	27.7	26.0	19.1	12.9	7.7
		MCWB	2.6	2.4	6.7	11.2	16.6	19.1	22.2	20.4	20.3	15.0	9.6	6.1
	10%	DB	2.3	2.9	7.8	15.2	22.9	25.0	27.8	26.3	23.9	16.8	11.1	5.0
		MCWB	0.6	1.0	4.7	9.6	16.0	18.1	21.0	19.9	19.2	12.7	8.2	3.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.4	10.2	14.6	16.8	22.0	23.6	26.2	24.1	23.9	19.7	15.0	10.6
		MCDB	11.7	12.0	20.4	23.6	28.6	29.3	32.4	29.2	29.1	22.1	17.3	12.5
	2%	WB	5.9	4.8	9.6	14.5	20.1	22.1	24.5	23.0	22.5	17.8	11.8	8.1
		MCDB	7.4	7.1	12.8	20.0	25.2	26.6	29.1	27.0	26.4	20.6	14.2	9.4
	5%	WB	2.8	2.9	6.9	12.1	18.8	20.8	23.2	22.1	21.2	15.4	10.3	6.0
		MCDB	4.4	4.9	9.7	16.6	22.9	24.5	27.6	25.8	24.9	18.1	13.0	7.5
	10%	WB	0.6	1.3	5.0	10.3	17.0	19.6	22.2	21.2	20.0	13.6	8.1	3.4
		MCDB	1.8	2.9	7.9	14.9	21.3	23.5	26.6	24.9	23.3	16.1	10.6	4.9
Mean Daily Temperature Range	5% DB	MDBR	9.0	8.9	9.0	10.5	11.8	10.7	10.9	10.5	10.8	8.7	7.7	7.2
		MCDBR	9.4	10.4	12.5	15.4	14.5	13.3	12.8	11.9	12.5	11.3	10.5	8.6
		MCWBR	8.8	8.4	8.5	9.0	7.3	6.5	5.8	5.8	6.9	7.2	8.0	8.3
	5% WB	MCDBR	9.4	10.2	11.1	13.9	11.7	10.2	10.7	9.8	11.4	9.8	9.7	8.5
		MCWBR	8.9	8.6	8.4	8.9	7.1	6.0	5.6	5.4	6.8	7.2	8.2	8.5
Clear-Sky Solar Irradiance	taub	0.275	0.290	0.328	0.357	0.381	0.433	0.440	0.417	0.374	0.335	0.311	0.287	
	taud	2.361	2.317	2.356	2.384	2.369	2.228	2.216	2.272	2.383	2.497	2.488	2.381	
	Ebn at Noon	845	896	905	904	888	839	828	834	844	838	800	787	
	Edh at Noon	75	95	106	113	119	137	137	125	103	80	68	68	
All-Sky Solar Radiation	RadAvg	1.43	2.32	3.49	4.31	5.05	5.50	5.58	4.98	3.99	2.32	1.44	1.02	
	RadStd	0.16	0.24	0.26	0.52	0.52	0.46	0.42	0.23	0.32	0.23	0.18	0.09	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (44 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+64	N/A

Nomenclature: See separate page